

DBMS

QUESTION BANK

(Frequently Asked Questions in VTU)

MODULE 1

Definition with example (study each for 3 to 4 marks):

- DBMS Catalog
 - Program-data independence
 - Cardinality of relationship/Degree of a relationship
 - Database Snapshot
 - Database System
 - Entity
 - Weak entity
 - Primary key
 - Foreign key
 - Multivalued attribute
 - Data dictionary
 - Participation types
- 1) Characteristics of Database approach (study for 10 marks)
 - 2) Advantages of Database management System (study for 10 marks)
 - 3) Difference of Database management System with File system (study for 10 marks)
 - 4) **3-schema Architecture** of DBMS with diagram/ with block diagram explain the architecture of a typical DBMS (study for 10 marks)
 - 5) Database System Environment (all the 3 parts) with block diagram
 - 6) Types of Database End Users with suitable example/ Explain different types of user friendly interfaces and types of user who typically use each. (study for 10 marks)
 - 7) Different Data Models
 - 8) Difference between database schema and database state
 - 9) Logical data independence & Physical data independence
 - 10) Structural constraints of a relationship-type with examples (study for 10 marks)
 - 11) ER Model/ER Diagram- all the 3 components of ER Diagram (1) Entity, Entity set, 2) different Attributes, 3) Relationship type, Relationship set, Cardinality, Mapping etc.)

*** From a given scenario, ER Diagram needs to be drawn. So, should know the notation of entity, different types of attributes, relation, cardinality mapping

MODULE 2

1. Explain domain integrity, key integrity, entity integrity and referential integrity constraint. Why is each considered important? Justify your answer with suitable examples. (study for 10 marks)
(or)
Explain with example basic constraints that can be specified when a database table/relation is created.
2. Explain the ER to Relational mapping algorithm with examples for each of the steps./ Summarize the steps involved in converting the ER constructs to corresponding relational tables. (study for 10 marks)
3. Explain following types of join operations in relational algebra with examples for each. (Must use relational algebra notations):
 - i. Theta join
 - ii. Equi join
 - iii. Natural join
 - iv. left outer join
 - v. right outer join
 - vi. full outer join
4. SQL Types with Syntax and Example: (i) DDL-Create, Alter, Drop, Truncate, (ii) DML-Insert, Update, Delete, (iii) DQL-Select (selecting all the columns, selecting specific columns based on where clause)
5. List and briefly explain the various attribute Data Types and Domains in SQL
6. Explain following SQL statements with Example for each:
 - i. INSERT
 - ii. DELETE
 - iii. UPDATE

***Also Queries related to create table,insert values, update table, alter table, select all/select specific column from a table.

7. Problems related to SQL Query statements (practice the problems as there in Text book/ shared samples in -> module wise note section)
8. Problems related to Relational Algebra operations(practice the problems as there in Text book/ shared samples in -> module wise note section)

MODULE 3

1. Explain the following constructs used in SQL with example (study each at least for 6-7 marks) :
 - (i) Views and their updateability / With an example, explain the concept of View in SQL
 - (ii) Assertion/ With an example, explain Specifying Constraints as Assertions in SQL
 - (iii) grouping—group by clause, having clause, order by clause
 - (iv) Nested queries
 - v) schema change statements
2. Differentiate between 'Where' and 'Having' clause in SQL with example.
3. Six clauses of SQL statement with example
4. Explain Cursor & its properties in Embedded SQL (study for 10 marks).
5. Explain JDBC drivers.
6. Explain the need of stored procedure with Example. Explain creating and calling stored procedure with Example
7. Explain Trigger concept with Example
8. Explain different Aggregate functions with example for each
9. What is SQLJ ? How does it differ from JDBC?
10. What is Dynamic SQL? How it is different from Embedded SQL
11. Draw and explain 3-tier Architecture and technology relevant to each tier. Write the advantages of 3- tier architecture/ With a neat diagram, Explain Three Tier Architecture Database Applications and briefly explain the advantages of Three tier Architecture [NOTE: It is **not same** as in 1st Module 3-tier architecture.. For this Question in each tier should show applications/ Technologies used]
12. Write a short note on
 - a)html forms b)Java scripts
 - c)CGI d)Application servers e)Servlets

MODULE 4

- 1) With suitable example, explain Informal Design Guidelines for Relation Schemas. (study for at least 10 marks)
- 2) With suitable example, explain First Normal Form, Second Normal Form, & Third Normal Form (study for at least 10 marks)
- 3) With suitable example, explain Properties of Relational Decompositions (study for at least 10 marks)
- 4) Explain insertion, deletion, and modification anomalies. Why are they considered bad?/ List explain different update anomalies. (study for at least 10 marks).
- 5) List & explain Armstrong's axioms
- 6) Explain the algorithm for Dependency-Preserving and Nonadditive (Lossless) Join Decomposition into 3NF Schemas (study for at least 10 marks)
- 7) Explain the algorithm for Nonadditive Join Decomposition into BCNF Schemas/ Define non-additive join property of a decomposition and write an algorithm for testing of non-additive join property. (study for at least 10 marks)
- 8) List and explain various functional dependencies: fully functional, partial functional, trivial dependency, transitive dependency, multivalued dependency.
- 9) Explain the normal forms that are based on fully functional dependency and transitive dependency.
- 10) Problems related to :
 - i. Finding closure of functional dependency
 - ii. Finding minimal cover of functional dependency
 - iii. Finding equivalent functional dependency

Sample problems for practice:

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- a) Consider the relation schema $R(A,B,C,D,E,F)$ and the functional dependencies $A \rightarrow B, C \rightarrow DF, AC \rightarrow E, D \rightarrow F$. What is the primary key of this relation R ? What is its highest normal form? Preserving the dependency, decompose R into third normal form.
 - b) Write the algorithm to find the minimal cover for a sets of FD's
Consider $R=\{A,B,C,D,E,F\}$.FD's $\{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ Find the irreducible cover for this set of FD's(minimal cover)
 - c) Given below are two sets of FD's for a relation $R(A,B,C,D,E)$.Are they equivalent?
 $F=\{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G=\{A \rightarrow CD, E \rightarrow AH\}$

MODULE 5

- 1) Why concurrency control and recovery are needed in DBMS? List & Explain the types of problem that may occur when 2 transactions run concurrently./ What are the anomalies that can occur due to interleaved execution? / What are the types of problems that may occur when 2 transactions run concurrently? (study for at least 10 marks)
- 2) With a neat diagram explain transition diagram of a transaction
- 3) Briefly discuss on the two-phase locking protocol/ 2PL used in concurrency control. How does it guarantee serializability. (study for at least 10 marks)
- 4) Explain different types of Locks used in concurrency control.
- 5) Explain Basic time stamping algorithm for Concurrency Control (study for 8 marks)
- 6) Explain multi version concurrency control protocols. (study for 8 marks)
- 7) Write short notes on the following (study for at least 7 -8 marks)
 - a. transaction rollback and cascading rollback.
 - b. shadow paging/ Explain how Shadow Paging helps to recover from transaction failure/ Write a short note on Shadow paging
 - c. NO-UNDO/REDO Recovery Based on Deferred Update
 - d. Recovery Techniques Based on Immediate Update
- 8) List and explain ACID Properties
- 9) Write a short Note on System Log
- 10) Briefly explain Transaction Support in SQL
- 11) Problems related to Testing conflict serializability of a Schedule using precedence graph (Practice the problems)
- 12) Check whether given schedule is serializable or not using precedence graph. Explain with algorithm.
 - a. S1:R1(X) R2(Z) R1(Z) R3(X) R3(Y) W1(X) W3(Y) R2(Y) W2(Z) W2(Y)
- 13) Consider the three transactions T1, T2, and T3, and the schedules S1 and S2 given below. Draw the serializability (precedence) graphs for S1 and S2, and state whether each schedule is serializable or not. If a schedule is serializable, write down the equivalent serial schedule(s). T1: r1 (X); r1 (Z); w1 (X); T2: r2 (Z); r2 (Y); w2 (Z); w2 (Y); T3: r3 (X); r3 (Y); w3 (Y); S1: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); w3 (Y); r2 (Y); w2 (Z); w2 (Y); S2: r1 (X); r2 (Z); r3 (X); r1 (Z); r2 (Y); r3 (Y); w1 (X); w2 (Z); w3 (Y); w2 (Y);

*** Refer sample problems & solutions in GCR->
Module wise note section